

EXAMPLES OF SOME PROPERTIES OF FUNCTORS

PETER J. HAINE

Exercise ([2, Exer. 1.5.xi]). Consider the functors $\mathbf{Ab} \hookrightarrow \mathbf{Grp}$ (inclusion), $\mathbf{Ring} \rightarrow \mathbf{Ab}$ (forgetting the multiplication), $(-)^{\times} : \mathbf{Ring} \rightarrow \mathbf{Grp}$ (taking the group of units), $\mathbf{Ring} \rightarrow \mathbf{Rng}$ (dropping the multiplicative unit), $\mathbf{Fld} \hookrightarrow \mathbf{Ring}$ (inclusion), $\mathbf{Mod}_R \rightarrow \mathbf{Ab}$ (forgetful). Determine which functors are full, which are faithful, and which are essentially surjective. Do any define an equivalence of categories?

In order to not state this many times, we remark that by definition it follows that an inclusion functor $C' \hookrightarrow C$ of a subcategory C' into a category C is faithful and injective on objects, and full just in case C' is a full subcategory. We use the characterization that functors which provide equivalences of categories are fully faithful and essentially surjective on objects to classify the functors that are equivalences, and say nothing more. Below is an enumerated list describing the functors in question.

- (a) *The inclusion functor $\mathbf{Ab} \hookrightarrow \mathbf{Grp}$ is fully faithful but not essentially surjective on objects, hence not an equivalence of categories.*

The inclusion is fully faithful since $\mathbf{Ab}$ is obviously a full subcategory of $\mathbf{Grp}$, however it is not essentially surjective on objects since not every group is abelian.

- (b) *The forgetful functor $U : \mathbf{Ring} \rightarrow \mathbf{Ab}$ is faithful, but neither full nor essentially surjective on objects, hence not an equivalence of categories.*

To see that U is faithful, notice that for any ring homomorphism $\phi : R \rightarrow R'$, is a homomorphism of the underlying group, but respects additional structure. Thus, in particular, for ring homomorphisms to be distinct, they must already be distinct as homomorphisms of underlying groups, which additionally respect the multiplicative structure. Hence the map of hom-sets is injective.

A basic fact from algebra is that there exists no ring homomorphisms $\mathbb{Z}/n \rightarrow \mathbb{Z}$. We know that $U(\mathbb{Z}/n)$ is the cyclic group $\mathbb{Z}/n$ and $U(\mathbb{Z})$ is the infinite cyclic group $\mathbb{Z}$, and there does exist a group homomorphism $\mathbb{Z}/n \rightarrow \mathbb{Z}$, the trivial homomorphism, hence U is not full.

To see that U is not essentially surjective on objects, we show that not every abelian group can have a ring structure on put on it. Fix a prime p and consider the *Prüfer p -group*

$$\mathbb{Z}(p^{\infty}) := \mathbb{Z}[1/p]/\mathbb{Z} \cong \left\{ e^{2\pi i m/p^n} \mid m, n \in \mathbb{N} \right\},$$

(so that $\mathbb{Z}(p^{\infty})$ can be regarded as the group of p^{th} -power roots of unity in the unit circle). Every element of $\mathbb{Z}(p^{\infty})$ has finite order, though there is no upper bound of the order of elements. Suppose, for the sake of contradiction, that there exists a unital ring R with $U(R) \cong \mathbb{Z}(p^{\infty})$. Then the multiplicative identity $1 \in R$ has finite additive order n , from which it follows that for all $r \in R$,

$$nr = n(1 \cdot r) = (n \cdot 1)r = 0 \cdot r = 0,$$

which shows that all elements of $U(R)$ have order at most n , contradicting the fact that $Z(p^\infty)$ has elements of arbitrarily large order.

- (c) *The functor $(-)^{\times} : \mathbf{Ring} \rightarrow \mathbf{Grp}$ is neither full, faithful, nor essentially surjective on objects, hence not an equivalence of categories*

To see that $(-)^{\times}$ is not full, notice that since Z is the initial ring, for any ring R there is a unique ring homomorphism $Z \rightarrow R$. However $Z^{\times} \cong Z/2$, and the group of units of F_p , for an odd prime p is $Z/(p-1)$. However, there are two group homomorphisms $Z/2 \rightarrow Z/(p-1)$, the trivial homomorphism and the homomorphism that sends the nontrivial element of $Z/2$ to the unique element of $Z/(p-1)$ of order 2, hence the map of hom-sets is not surjective.

To see that $(-)^{\times}$ is not faithful, suppose k a field and consider the polynomial ring $k[t]$. It is a well-known fact that the automorphisms of $k[t]$ that fix k are the affine maps $t \mapsto at + b$, where $a \in k^{\times}$. Since the group of units of $k[t]$ is simply $k^{\times}$, any two automorphisms fixing k induce the identity homomorphism on $k^{\times}$, so the map

$$\mathbf{Ring}(k[t], k[t]) \rightarrow \mathbf{Grp}(k^{\times}, k^{\times})$$

is not injective in general as there are usually many affine transformations of $k[t]$ (even for F_2).

A paper of Pearson and Schneider from 1970 shows that not every cyclic group is isomorphic to the group of units of some ring [1, Cor. 3], though this is nontrivial. In particular, the cyclic group $Z/5$ is not isomorphic to $R^{\times}$ for any ring R . This proves that $(-)^{\times}$ is not essentially surjective on objects

- (d) *The inclusion functor $\mathbf{Ring} \hookrightarrow \mathbf{Rng}$ obtained by dropping the multiplicative unit axiom is faithful but neither full nor essentially surjective on objects, hence not an equivalence of categories.*

The inclusion is not full since the zero homomorphism is not a homomorphism in $\mathbf{Ring}$ but is a homomorphism in $\mathbf{Rng}$: since Z is the initial ring, there is one homomorphism $Z \rightarrow R$ in $\mathbf{Ring}$, but there are two homomorphisms $Z \rightarrow R$ in $\mathbf{Rng}$. It is not essentially surjective on objects since a ring with no multiplicative unit clearly is not isomorphic to a ring with a multiplicative unit.

- (e) *The inclusion functor $\mathbf{Fld} \hookrightarrow \mathbf{Ring}$ is fully faithful, but not essentially surjective on objects, hence not an equivalence of categories.*

Since a field homomorphism is simply a ring homomorphism between fields, the category $\mathbf{Fld}$ forms a full subcategory of $\mathbf{Ring}$. Since not every ring is isomorphic to a field, the inclusion is not essentially surjective on objects.

- (f) *In the forgetful functor $U : \mathbf{Mod}_R \rightarrow \mathbf{Ab}$ is faithful and essentially surjective on objects, but need not be full—moreover, the fullness of U depends on the underlying ring R . Hence, in general, the forgetful functor is not an equivalence of categories.*

To see that U is faithful, notice that for any R -module homomorphism $M \rightarrow M'$, is a homomorphism of the underlying group, but respects additional structure. Thus, in particular, for R -module homomorphisms to be distinct, they must already be distinct as homomorphisms of underlying groups and in addition respect the module structure. Hence the map of Hom-sets is injective.

To see that U is essentially surjective on objects, it suffices to notice that given any ring R and abelian group A , there is a trivial R -module A_R given by setting $ra = a$ for $r \in R$ and $a \in A$. Then $U(A_R) = A$.

Now notice that U is not necessarily full. Treating a ring R as an R -module, since an endomorphism $\phi : R \rightarrow R$ is uniquely determined by $\phi(1)$ we have that $\text{End}_R(R) \cong R$. In general, however, group homomorphisms are only determined by the image of

one group element when the group is free, so in general the endomorphisms ring of the underlying group is not isomorphic to R , though obviously this depends on the underlying ring R . When $R = \mathbf{Z}$, this is true, and, moreover, since abelian groups are precisely $\mathbf{Z}$ -modules, it is obvious that the “forgetful” functor $U: \mathbf{Mod}_{\mathbf{Z}} \rightarrow \mathbf{Ab}$ is, in particular, an equivalence of categories.

REFERENCES

1. K. R. Pearson and J. E. Schneider, *Rings with a cyclic group of units*, Journal of Algebra **16** (1970), 243–251.
2. Emily Riehl, *Category theory in context*, Aurora: Dover Modern Math Originals, Dover Publications, 2016.